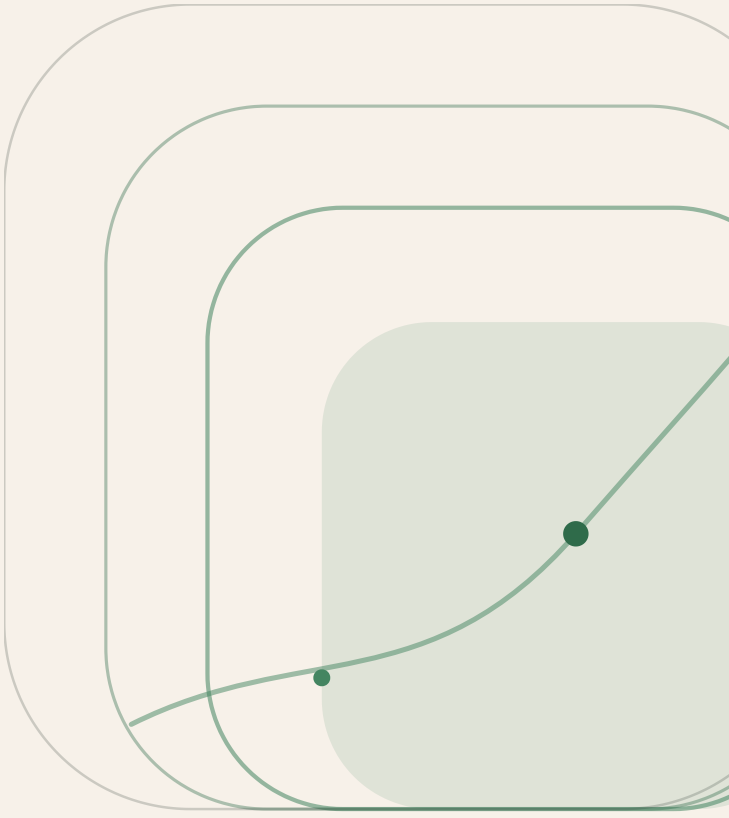
Greecon
SH.P.K

GUIDE 03 · FOR THE PROJECT MANAGER AND THE ELECTRICIAN

Hardware Reference

Every part of a kit: the box, the converter, the meters, the input modules, the level, pressure, flow, soil and climate sensors, the cellular router, power, the UPS, the enclosure, labels, tools and spares.

FOR	The project manager who buys and fits hardware, and the electrician
PRINT	Once, to keep with the spares kit
COVERS	Specifications, planning estimates, settings, fitting, checks and the mistakes to avoid, part by part



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Hardware reference

This part describes every kind of hardware in a Greecon kit: what it is, what it must be, what it costs, how it is set up, how it is fitted, how to check it and what goes wrong. It is written from the hardware and wiring document (docs/launch/03) and is the place to look when a pack says “follow the maker’s manual”. The four site packs (PDFs 04 to 07) use these parts in four combinations.

TIP

Where this part is silent, the maker’s manual decides. The drawings show how the parts connect, not terminal numbers: terminal numbers differ between makes and versions. Every drawing says so. Part numbers named here are examples of a kind of part; the specification is what matters. Prices are planning estimates from public shop prices seen in October 2026, not quotes: replace them with real quotes in the first week and keep the quotes in the shared drive.

Each part also has a sheet of its own, in the Qualified Hardware Catalogue (PDF 10) and as a PDF by itself (HW01 to HW17): how it qualifies, what it does at each of the four kinds of site, its acceptance test and a buying record.

4.1 Rules of thumb

The aim is the cheapest way to run many sites for many years: the same few parts everywhere, prepared at the office, installed in an afternoon, and replaceable by anyone because nothing about a box is special. The parts that talk to the platform are **approved models**: the two Eastron meters (SDM630 and SDM230) and the two Waveshare modules (Modbus RTU Analog Input 8CH and Modbus RTU IO 8CH). Greecon knows their registers, their serial settings and the way to recognise them on a network, so nobody on site reads a datasheet or types a register. The rest of a kit (the box, the converter, the supply, the transmitters, the enclosure) is bought by specification and not by brand. Greecon speaks two standard languages, Modbus over the network and a wire signal of 4 to 20 mA read through a Modbus module, so a part that is not an approved model can be swapped when a maker stops selling it.

1. **One of each first.** Buy one full kit and bench-test it (the rehearsal, the Project Manager Field Manual (PDF 02), section 3.11) before buying five. Change the specification, not the kit, when something fails.
2. **Read-only, always.** No relay output, no control wire, no write. Everything here only measures. (The pump contact module has relay outputs: they are never used.)
3. **Approved models for what is read over the cable, a published register table for anything else.** A kit is made of the approved models and no others. A soil probe, a weather sensor or a battery is not an approved model and is mapped by hand: if the maker will not give the register table, do not buy it, because it cannot be mapped honestly.
4. **Everything powered from one clean place,** on a UPS, in one enclosure, labelled.
5. **Pre-stage at the office:** the box has its software installed and tested, every device has its number and its speed, and the converter is set, before the kit leaves.

6. **Same settings everywhere:** unit numbers (grid meter 1, solar meter 2, 4 to 20 mA water module 3, load or pump meter 4, pump contact module 5, soil probe 6, weather sensor 7), speed 9600, 8 data bits, no parity, 1 stop bit, the converter on port 502. A new site then looks like the last one.

STOP

Greecon only reads, and only an electrician works on mains. Nothing in this part tells anyone to work on a live conductor. Everything inside a distribution board, every current clamp and voltage tap, and the pump's run contact are the electrician's work, done under the customer's lock-out procedure. The project manager does the low-voltage and software side. Greecon is not a safety system: the float switches, relief valves and trips already in the customer's equipment stay the safety layer.

4.2 The standard kits

Four kits cover the four kinds of site. Costs are planning estimates in EUR, excluding VAT and installation labour. A part marked optional is not always fitted. The meters and the two modules are approved models: the model named is the model bought. The first table is every part once; the tables after it are the four kits, with quantities.

Every part, what it does and which kits use it

Kits in brackets use the part only sometimes.

Part	What it does	Kits	Planning estimate (EUR)
Box	Reads the meters and sensors and reports to Greecon. If the internet is down it keeps about three days of readings (for a site of up to about thirteen devices) and sends them in order when the internet returns.	H S F W	180 to 300
Converter	Puts the meters' cable (RS-485) on the network, so that the box can reach every meter and module on it. It must be in Modbus TCP to RTU gateway mode, and is set at the office (9600 baud, 8 data bits, no parity, 1 stop bit, TCP port 502).	H S F W	30 to 60 (basic) or 250 and up (industrial)
Three-phase meter	Measures electrical power on three phases. One at the grid connection (unit 1) gives both Grid import and Grid export; one on the inverter's AC output (unit 2) gives Solar production; a third, if wanted, on the site's own consumption (unit 4). Fitted by the electrician.	H S	90 to 180 each
4 to 20 mA input module	Reads the 4 to 20 mA signals of the level, pressure and flow transmitters (inputs 1, 2 and 3) and answers the box on the same RS-485 cable as the meters, as unit 3. It is left in its default 0 to 20 mA mode, so that a wire that has come off reads zero and Greecon can say so.	H F W	30 to 60
Pump contact module	Reports whether the pump is running, from a spare contact of its starter (a dry contact on input 1), as unit 5. It only reports; it never switches anything.	H (F) W	30 to 60
Level transmitter	Measures how full the tank is. It hangs at the bottom of the tank.	H F W	120 to 400
Pressure transmitter	Measures the pressure in the pipe, through an isolating valve so that it can be removed without draining.	H F W	40 to 150

Part	What it does	Kits	Planning estimate (EUR)
Flow meter	Measures the flow in the pipe. Without it (or without the pump's running contact) the alert for a pump that runs with no water moving cannot be raised.	(H) (F) (W)	150 to 500
24 V DIN power supply	Powers the two modules, the loop transmitters and the converter from one place.	H S F W	25 to 50
UPS	Keeps the box and the converter up through short power cuts and long enough to say that the grid is down.	H S W	60 to 150
Enclosure and fittings	Holds the converter, the modules, the supply, the fuses, the terminal blocks and, if there is room, the box.	H S F W	80 to 150
Cable and labels	The RS-485 data cable (shielded twisted pair), network patch cords, ferrules and labels.	H S F W	40 to 80
Cellular router	Gives the box and the converter an internet connection where the site has no usable one: a 4G router with a data SIM.	(H) (S) F (W)	150 to 200
Single-phase meter	Measures the electricity of one pump or a small load (single-phase), on its own feed (unit 4).	(W)	35 to 60
Soil moisture probe	Measures soil moisture at the probe's depth. Its depth and place are written down, because soil readings mean little without them. Unit 6, mapped by hand from the maker's register table.	F	40 to 120 each
Temperature and humidity sensor	Measures air temperature and humidity at the field. Unit 7, mapped by hand from the maker's register table.	F	20 to 80

Kit H: hotel or guesthouse (water, solar, grid)

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Box**	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300
☐	**Converter**	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)
☐	**Three-phase meter**	Eastron SDM630 Modbus. The MCT version comes with three clamp-on current sensors for up to 100 A a phase; for more than 100 A, the ordinary SDM630 with 5 A current transformers. *One at the grid connection, one on the inverter's AC output.*	2	90 to 180 each
☐	**4 to 20 mA input module**	Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60
☐	**Pump contact module**	Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used. *Leave it out if the starter has no spare running contact.*	1	30 to 60

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Level transmitter**	Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400
☐	**Pressure transmitter**	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150
☐	**Flow meter** *(optional)*	4 to 20 mA output. Needed for "the pump is running but no water is flowing". *Needed for the no-flow alert.*	0 or 1	150 to 500
☐	**24 V DIN power supply**	30 to 60 W	1	25 to 50
☐	**UPS**	A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150
☐	**Enclosure and fittings**	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150
☐	**Cable and labels**	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80
☐	**Cellular router** *(optional)*	4G, DIN rail, data SIM. *Only without usable internet.*	0 or 1	150 to 200

About EUR 800 to 1,800; typically 1,100 to 1,300. Add the electrician's time (half a day to a day) and yours.

Kit S: solar plant with grid connection

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Box**	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300
☐	**Converter**	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)
☐	**Three-phase meter**	Eastron SDM630 Modbus. The MCT version comes with three clamp-on current sensors for up to 100 A a phase; for more than 100 A, the ordinary SDM630 with 5 A current transformers. *Grid and solar; a third on the site's own consumption if the owner wants it measured.*	2, or 3	90 to 180 each
☐	**24 V DIN power supply**	30 to 60 W	1	25 to 50
☐	**UPS**	A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150
☐	**Enclosure and fittings**	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150
☐	**Cable and labels**	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80
☐	**Cellular router** *(optional)*	4G, DIN rail, data SIM. *Only without usable internet; not in the hardware guide's list for this kit, see the network rules.*	0 or 1	150 to 200

For a battery add nothing but the maker's Modbus register table (the battery speaks Modbus itself on the site's network or on its own RS-485 port; if the maker will not publish the table, leave the battery out). **About EUR 600 to 1,200.**

Kit F: farm or greenhouse with irrigation

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Box**	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300
☐	**Converter**	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)
☐	**4 to 20 mA input module**	Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60
☐	**Level transmitter**	Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Pressure transmitter**	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150
☐	**24 V DIN power supply**	30 to 60 W	1	25 to 50
☐	**Enclosure and fittings**	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150
☐	**Cable and labels**	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80
☐	**Cellular router**	4G, DIN rail, data SIM. *The likelier case in a field.*	0 or 1	150 to 200
☐	**Soil moisture probe**	RS-485 Modbus, with a published register table, one for each zone	1 for each zone	40 to 120 each
☐	**Temperature and humidity sensor**	RS-485 Modbus, with a published register table	1	20 to 80
☐	**Pump contact module** *(optional)*	Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used. *If there is a pump to watch.*	0 or 1	30 to 60
☐	**Flow meter** *(optional)*	4 to 20 mA output. Needed for "the pump is running but no water is flowing". *If there is a pump to watch.*	0 or 1	150 to 500

The standard list for this kit has no UPS row, although rule 4 in section 4.1 says everything is powered on a UPS: decide at the survey whether this site gets one, and write the decision down. A cellular router is the likelier case in a field. **About EUR 700 to 1,500.**

Kit W: water tank and pump station

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Box**	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300
☐	**Converter**	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)
☐	**4 to 20 mA input module**	Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60
☐	**Pump contact module**	Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used. *Leave it out if the starter has no spare running contact.*	1	30 to 60
☐	**Level transmitter**	Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Pressure transmitter**	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150
☐	**Flow meter** *(optional)*	4 to 20 mA output. Needed for "the pump is running but no water is flowing". *Needed for the no-flow alert.*	0 or 1	150 to 500
☐	**24 V DIN power supply**	30 to 60 W	1	25 to 50
☐	**UPS**	A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS	1	60 to 150
☐	**Enclosure and fittings**	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150
☐	**Cable and labels**	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80
☐	**Cellular router** *(optional)*	4G, DIN rail, data SIM. *Only without usable internet.*	0 or 1	150 to 200
☐	**Single-phase meter** *(optional)*	Eastron SDM230, on the pump's feed, to measure its electricity. *To measure the pump's electricity.*	0 or 1	35 to 60

About EUR 650 to 1,300.

The spares kit (one for each project manager)

One of each, so that a failed part is swapped on the day:

- ☐ A converter
- ☐ A 4 to 20 mA module
- ☐ A pump contact module
- ☐ A meter
- ☐ A 24 V supply
- ☐ A transmitter (pressure)
- ☐ A spare box, pre-installed, with no code, ready to take one
- ☐ Network cables
- ☐ A loop calibrator
- ☐ A multimeter
- ☐ Fuses
- ☐ A spare SIM

4.3 The box

The box runs one small program and holds a queue of readings. It needs very little, and it must survive power cuts, which is the hard part.

WHAT IT DOES	Reads the meters and sensors and reports to Greecon. If the internet is down it keeps about three days of readings (for a site of up to about thirteen devices) and sends them in order when the internet returns.
SPECIFICATION	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input
PLANNING ESTIMATE (EUR)	180 to 300. Not a quote.
USED IN	Kits H, S, F, W

What the box needs

Needs	Why
Linux on a processor of the x86-64, 64-bit ARM or 32-bit ARM kind (the installer checks)	The software is a Node.js program
Debian 12, Ubuntu 22.04 or later, or Raspberry Pi OS (64-bit), with systemd	What the installer supports
4 GB of memory or more, 32 GB of storage or more	The program asks for about 400 MB; the queue is on disk
A real disk (SSD or eMMC), not an SD card	An SD card is the commonest reason these boxes fail: they corrupt on power cuts and wear out under constant writing
A real-time clock with a battery, and a BIOS setting of “power on after power loss” (often “Restore on AC Power Loss: Power On”)	After a cut the box must come back by itself, with the right time
9 to 36 V DC input, or a plain 12 V one	So that it can sit behind a DC UPS and share the 24 V supply
No fan, 0 to 50 °C or better	A plant room is hot and dusty
One Ethernet port (two is nicer)	One to the converter’s network, one spare

The kit list buys 8 GB and 128 GB or more; this table is the least the software will run on.

Options

Option	When	Planning estimate (EUR)	Note
Fanless x86 mini PC (an Intel N100-class machine)	The default. Most sites.	180 to 300	Check the real-time-clock battery and the power-on setting in the BIOS before buying many.
Raspberry Pi 5 (4 or 8 GB)	A bench box, or a small site with a proper SSD on NVMe or USB 3 and an RTC battery. Never an SD card.	120 to 200 with SSD and case	Starts by itself when power returns.
Industrial PC (Advantech UNO, OnLogic, Seeed reTerminal DM class)	Hot, dusty or vibrating sites; customers who insist on industrial parts	450 to 900	The same software runs on it. Not needed to begin.

Preparing the operating system (at the office, twenty minutes)

1. Install Debian 12 from the “netinst” image: only “SSH server” and “standard system utilities”. No desktop.
2. One user with a long password (in the password manager, against the site’s name). Host name `greecon-<site>`.
3. Time zone Europe/Tirane; `timedatectl set-ntp true`.
4. Automatic security updates: `apt-get install -y unattended-upgrades` and answer yes when it asks.
5. BIOS: power on after power loss; no sleep; boot from the disk only.
6. Then paste the install line (the Project Manager Field Manual (PDF 02), section 3.6).

What the installer changes

So that nobody wonders: `/opt/greecon` (the software), `/etc/greecon` (the box’s credential, readable by the service only), `/var/lib/greecon-edge` (what it remembers), one service called `greecon-edge`, one user called `greecon`, and one command, `greecon-edge`. Nothing else.

What the box does by itself

- It restarts if it fails.
- It keeps about three days of readings if the internet is down, for a site of up to about thirteen devices (a site with more devices keeps fewer days), and sends them in order when it returns. A long outage keeps one reading a minute for each reading, so that three days fit.
- It takes new versions of its software from the platform and goes back to the version before if a new one fails to start three times.
- It reports every 30 seconds so that Greecon knows it is alive.

Commands the project manager uses

```
greecon-edge status                       how the box is: heard by Greecon? each device ok? the
queue
journalctl -u greecon-edge -n 50 --no-pager   what it has been saying
greecon-edge find                         look at the network and say what answers (reads only)
greecon-edge modbus-test --host 192.168.1.50 --unit 1 --type input --address 52 --data-type
float32
sudo systemctl restart greecon-edge       restart it
```

CHECK

Before the box leaves the office: the install line has run to “Done. Greecon can see this box.”, the setup page shows the install step done with the box’s name and “Heard from N seconds ago”, the box is labelled with the site name and the date, and the BIOS power-on setting has been tried by pulling the power and restoring it.

4.4 The converter and the meter cable

The meters and modules speak Modbus RTU on a two-wire cable (RS-485). The box speaks Modbus TCP on the network. The converter sits between them.

WHAT IT DOES	Puts the meters' cable (RS-485) on the network, so that the box can reach every meter and module on it. It must be in Modbus TCP to RTU gateway mode, and is set at the office (9600 baud, 8 data bits, no parity, 1 stop bit, TCP port 502).
SPECIFICATION	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode
PLANNING ESTIMATE (EUR)	30 to 60 (basic) or 250 and up (industrial). Not a quote.
USED IN	Kits H, S, F, W

STOP

The setting that matters most: the converter must be in **Modbus TCP to RTU gateway mode** (sometimes called "Modbus gateway", "Modbus TCP server" or "protocol conversion"). Not "transparent" or "raw TCP" mode, and not "Modbus RTU over TCP": in those, the box's requests arrive on the cable as garbage and nothing answers.

The settings in the converter's web page

Greecon sets them at the office, before the kit leaves. They are here in case the converter ever has to be set again.

Setting	Value
Mode	Modbus TCP to RTU (gateway)
Serial: speed (baud)	9600 (the same as every meter on the cable)
Serial: data bits, parity, stop bits	8, none, 1
TCP: port	502
Address	A fixed address on the site's network (a reservation in the router, or set in the converter)

The box keeps one TCP connection per converter and asks one thing at a time, so any converter that does the above will do.

Wiring RS-485

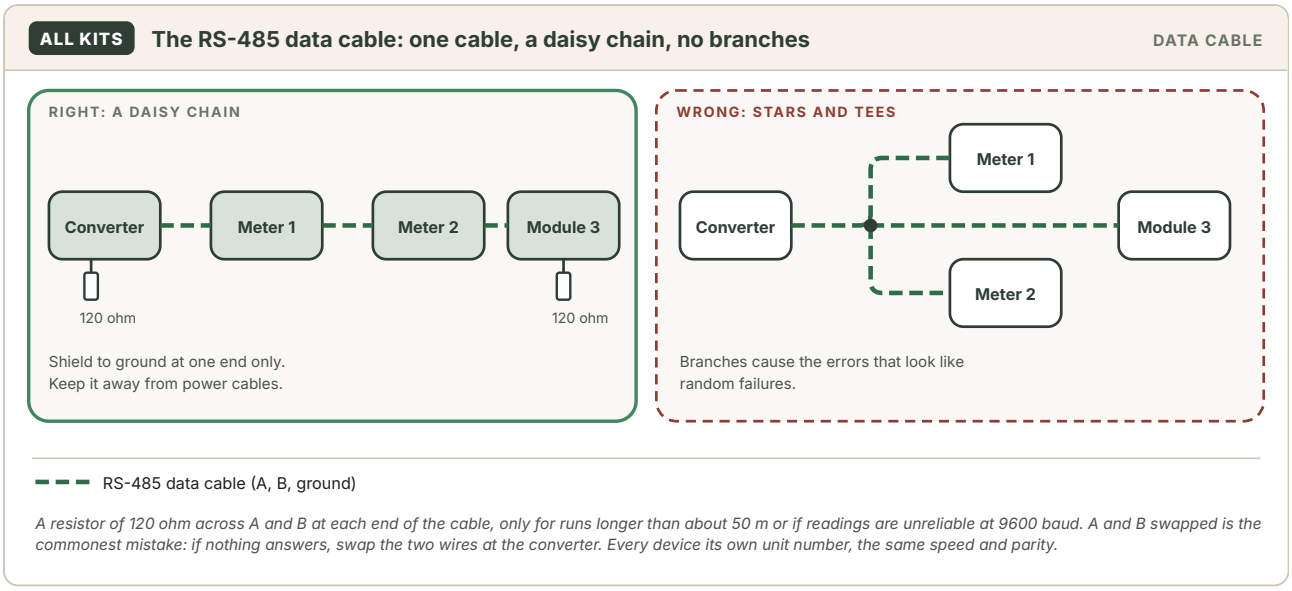


Figure 4-1. The RS-485 data cable: one cable and a daisy chain, never branches.

- **One cable, daisy chain,** from the converter to the first device, on to the next, and so on. No branches: stars and tees cause the errors that look like random failures.
- Use **shielded twisted pair,** the shield joined to ground at one end only. Keep it away from power cables.
- **A and B swapped** is the commonest mistake. If nothing answers, swap the two wires at the converter.
- **Every device its own unit number,** the same speed and parity. The meters and the Waveshare modules are delivered as number 1, so each is numbered at the office before it joins the cable: two with the same number would answer over each other.
- A 120 ohm resistor across A and B at each end of the cable is only for runs longer than about 50 m, or if readings are unreliable at 9600 baud.

The number each device has on the cable

Device	Number	How it is set, and when
Grid meter (Eastron SDM630)	1	On its display, at the office
Solar meter (Eastron SDM630)	2	On its display, at the office
4 to 20 mA module (Waveshare Modbus RTU Analog Input 8CH)	3	With the maker’s setting tool, at the office
Load meter (SDM630) or pump meter (SDM230)	4	On its display, at the office
Pump contact module (Waveshare Modbus RTU IO 8CH)	5	With the maker’s setting tool, at the office
Soil probe	6	The maker’s way, before it is buried
Weather sensor	7	The maker’s way, before it is mounted
A battery	By hand, for example 8	From the battery maker’s manual

Every device on one cable has the same speed: 9600 baud, 8 data bits, no parity, 1 stop bit. On site you check the label on each device against the install card and do not change it.

Cable and labels

WHAT IT DOES	The RS-485 data cable (shielded twisted pair), network patch cords, ferrules and labels.
SPECIFICATION	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels
PLANNING ESTIMATE (EUR)	40 to 80. Not a quote.
USED IN	Kits H, S, F, W

Label the unit number on the converter and on every device, the wire number at both ends, and the site name and date on a sticker inside the lid of the enclosure.

4.5 The meters

Eastron SDM630 (three-phase) at the grid connection, where the supply enters, and on the inverter's AC output. Eastron SDM230 (single-phase) for one pump or a small load.

Three-phase meter

WHAT IT DOES	Measures electrical power on three phases. One at the grid connection (unit 1) gives both Grid import and Grid export; one on the inverter's AC output (unit 2) gives Solar production; a third, if wanted, on the site's own consumption (unit 4). Fitted by the electrician.
SPECIFICATION	Eastron SDM630 Modbus. The MCT version comes with three clamp-on current sensors for up to 100 A a phase; for more than 100 A, the ordinary SDM630 with 5 A current transformers
PLANNING ESTIMATE (EUR)	90 to 180 each. Not a quote.
USED IN	Kits H, S

Single-phase meter

WHAT IT DOES	Measures the electricity of one pump or a small load (single-phase), on its own feed (unit 4).
SPECIFICATION	Eastron SDM230, on the pump's feed, to measure its electricity
PLANNING ESTIMATE (EUR)	35 to 60. Not a quote.
USED IN	Optional in kit W

The battery (kit S, optional)

A battery is not bought from Greecon's list: it is whatever the customer has, and it is not an approved model. Greecon can read it only if the maker publishes the battery's Modbus register table. The battery speaks Modbus itself, on the site's network (its own address) or on its own RS-485 port (then it joins the cable with a

number of its own, by hand, for example 8). The box cannot recognise it on a network, so its address is entered by hand. If the maker will not publish the table, leave the battery out: do not guess registers. Two readings are mapped by hand from the maker's table: Battery charge (percent) and Battery power (kW, positive while charging). If there is no battery, remove the template's battery device (the project manager's step in the pack for kit S).

What Greecon reads

Total active power: register 52 on the SDM630 (address counted from zero, as a manual's 30053) and register 12 on the SDM230, 32-bit floating point, high half first, in watts; Greecon shows it in kW. Positive power is energy coming in, negative is energy sent out. That is why one grid meter gives both Grid import and Grid export, and why a clamp fitted the wrong way round swaps them (the direction check, setup step 9).

The Eastron meters in the templates already have their registers filled in. The unit scale applied is only watts to kilowatts. Greecon reads what the meter shows.

Settings made at the office

These differ between makes and versions, so they are read from the meter's own display, never assumed. Greecon sets them at the office, before the kit leaves; on site you only check them.

Setting	Value
Unit number	A different one on each meter: grid meter 1, solar meter 2, load or pump meter 4. Set on the meter's display
Speed	SDM630: 9600 as delivered, so check it. SDM230: depends on the version (the V1.6 manual says 2400); set it to 9600 on the meter's display, on the Baud Rate page
Parity and stop bits	None, and as the manual says (the converter uses 8 data bits, no parity, 1 stop bit)
CT ratio	On a meter with 5 A current transformers, set the ratio in the meter's menu (for example 250 for a 250/5 A transformer) so that the meter shows the real power. It depends on the transformers fitted, so it is set on site. Greecon reads what the meter shows

A meter's display works only when its voltage terminals have power, and powering a meter on a bench is mains work: the electrician makes up the office's bench lead, and the project manager never wires mains.

Fit the model the install card lists. Greecon reads an SDM630 by its register 52 and an SDM230 by its register 12. If the box finds a single-phase SDM230 where a three-phase SDM630 was planned, **Find equipment** says so, and reads it as an SDM230 once you press **Use this address**. An SDM630 fitted where an SDM230 was planned is read as an SDM230 and shows one phase's power only: **Find equipment** says it may be a three-phase meter, and the commissioning comparison with the meter's own display finds it too.

Fitting: the electrician’s work

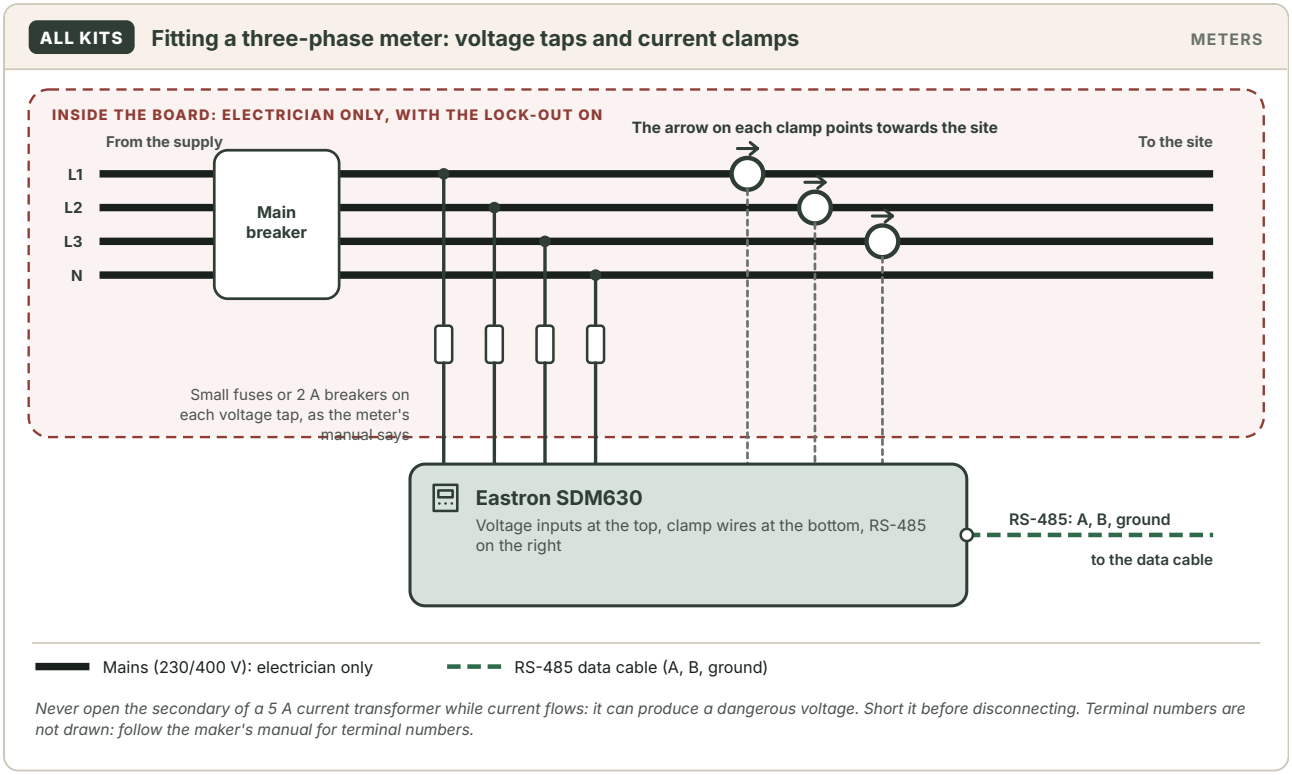


Figure 4-2. Fitting a three-phase meter: fused voltage taps and three current clamps, arrows towards the site. The electrician's work.

STOP

Inside a distribution board is the electrician’s work, with the lock-out on. The project manager watches, photographs and writes it down.

- **Clamps point towards the site.** On the grid meter this makes import positive. On the solar meter, the clamps point away from the inverter, towards the board (towards the site), so that production is read. The platform takes the size of the power for solar, so a reversed solar clamp does not read zero, but fit it the right way anyway.
- **Never open the secondary of a 5 A current transformer while current flows:** it can produce a dangerous voltage. Short the secondary before disconnecting. This is the electrician’s rule; it is repeated here because it is the one that hurts.
- Voltage taps have small fuses or 2 A breakers on each tap, as the meter’s manual says.
- **Photograph** the wiring, the label (model and serial number) and the display with its settings.

CHECK

Check on the day: with the sun down, switch on a known load: grid import must go up. At night solar reads zero or a few watts. At midday the solar meter agrees with the inverter’s own display within about 5 percent.

4.6 Water: tank, pressure, flow, pump

Two modules read the water side, both on the same RS-485 cable as the meters, and both approved models: the **4 to 20 mA module** (unit number 3) reads the level, pressure and flow transmitters, and the **pump contact module** (unit number 5) reads whether the pump is running. Greecon sets their numbers at the office; on site you check the labels.

The 4 to 20 mA module

WHAT IT DOES	Reads the 4 to 20 mA signals of the level, pressure and flow transmitters (inputs 1, 2 and 3) and answers the box on the same RS-485 cable as the meters, as unit 3. It is left in its default 0 to 20 mA mode, so that a wire that has come off reads zero and Greecon can say so.
SPECIFICATION	Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC
PLANNING ESTIMATE (EUR)	30 to 60. Not a quote.
USED IN	Kits H, F, W

Its inputs are left in their default 0 to 20 mA mode, on purpose. A 4 to 20 mA transmitter then reads **4000 to 20000 microamps**, and a wire that has come off reads **0**, which is below the 4 mA of an empty tank, so Greecon can see it and say so. (In the module's own 4 to 20 mA mode a broken wire might read like an empty tank.) Use the version that reads current, not the one that reads voltage.

Each transmitter has its own input: the level transmitter on **input 1**, the pressure transmitter on **input 2** and the flow meter on **input 3**. Greecon asks for the range printed on each label (below) and works out the register and the scale: nobody types a register or a conversion.

A 4 to 20 mA loop

A two-wire transmitter takes its power from the loop, so the loop needs a 24 V supply in series with the module's input. The electrician wires it as the module's manual draws it. The maker does not say that this module supplies the loop, so the loop has a supply of its own.

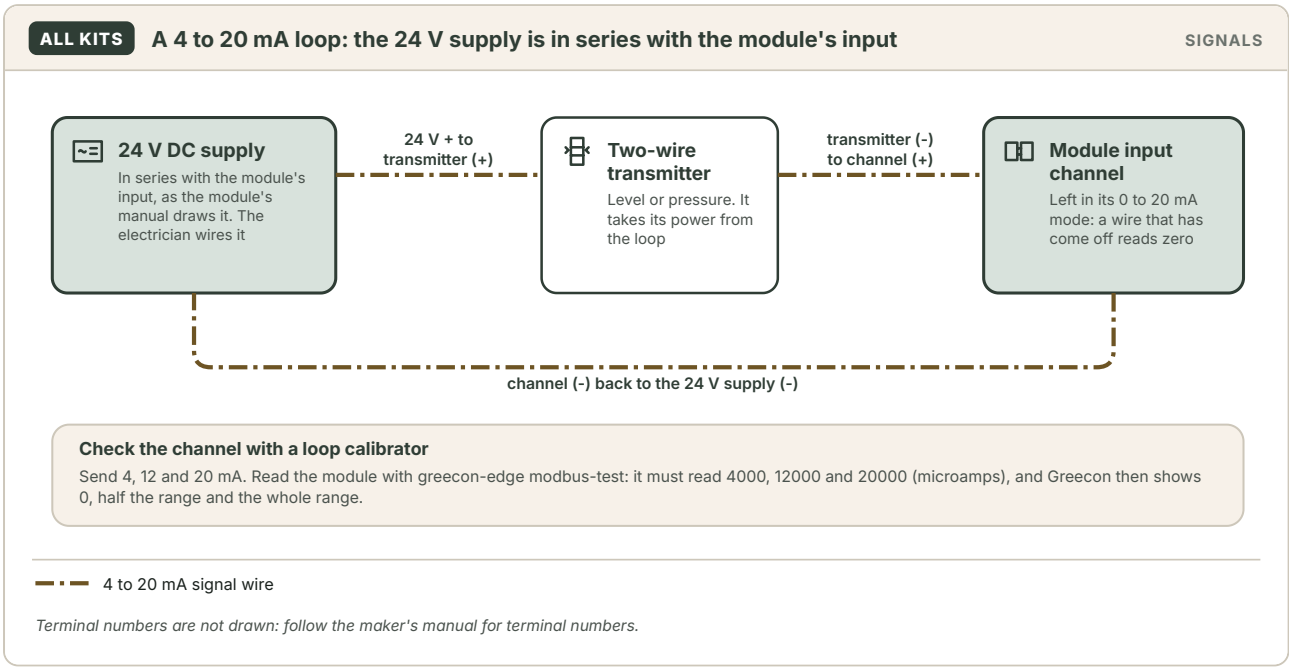


Figure 4-3. A 4 to 20 mA loop: the 24 V supply in series with the module's input.

Ranges from the labels

The range of a transmitter is printed on its label, and the setup list asks for nothing else, in the label's own words:

- **Level transmitter:** the range on the label (a 0 to 5 m transmitter: 5), its unit (metres, or bar for a pressure sensor hung at the bottom of the tank: a bar holds up about 10.2 m of water) and how tall the tank is inside, from the bottom to the top of the water when it is full (say 4, in metres). Greecon shows a percentage of the tank: 0 when empty, 100 when full. A 0 to 5 m transmitter in a tank 4 m tall reads 100 at 16.8 mA and 125 at the top of its range (5 ÷ 4 × 100). One shorter than the tank is refused: it cannot see the top of the water.
- **Pressure transmitter:** the top of the range in bar (0 to 10 bar is usual).
- **Flow meter:** the top of the range in litres a minute.
- A transmitter the site does not have: tick **Not fitted at this site**. Greecon asks first: the reading and its history are removed.

Check the module with a loop calibrator (a hand-held that sends 4, 12 and 20 mA in place of a transmitter). Read the module from the box (address 0 is input 1, 1 is input 2, 2 is input 3):

```
greecon-edge modbus-test --host 192.168.1.50 --unit 3 --type input --address 0
```

It must read 4000, 12000 and 20000, and Greecon then shows 0, half the range and the whole range. On the tank's input the whole range reads more than 100 when the transmitter is longer than the tank (125 in the example), and Greecon may call that impossible while you test: that is the test, not a fault.

The pump contact module

WHAT IT DOES	Reports whether the pump is running, from a spare contact of its starter (a dry contact on input 1), as unit 5. It only reports; it never switches anything.
SPECIFICATION	Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used
PLANNING ESTIMATE (EUR)	30 to 60. Not a quote.
USED IN	Kits H, W; optional in kit F

A spare auxiliary contact of the pump's starter is wired by the electrician to **input 1** as a dry contact, as the module's manual draws it (the module's DI COM is left unconnected). Never to a control circuit, and **never to the module's relay outputs**: Greecon only reads. **Check it by eye:** with the pump running, Pump running must read 1; stopped, 0. If it reads the other way round, the contact is inverted: the electrician uses the other contact of the starter. The module is optional: a pump whose starter has no spare running contact has none, the site then removes it from the list, and the handover says that "the pump is running but no water is flowing" cannot be raised.

The level transmitter

WHAT IT DOES	Measures how full the tank is. It hangs at the bottom of the tank.
SPECIFICATION	Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)
PLANNING ESTIMATE (EUR)	120 to 400. Not a quote.
USED IN	Kits H, F, W

- The transmitter hangs at the bottom of the tank, so that zero is empty, its vented cable fixed so that it cannot move.
- **Only a transmitter with wetted parts approved for drinking water goes into a drinking-water tank.** A cheap sensor of unknown material is not worth a contaminated tank. If unsure, use a non-contact ultrasonic sensor above the water instead.
- The electrician or the customer's plumber fits it. If the tank must be entered, the customer's own staff do that and the project manager does not: if the only place for the sensor is inside, say it is not possible and write why.
- **In Greecon** give the range on the label and how tall the tank is inside (from the bottom, where the transmitter hangs, to the top of the water when it is full). Greecon works out the percentage: a 0 to 5 m transmitter in a tank 4 m tall reads 100 when the tank is full (16.8 mA), and 125 at the top of its range. Wired to input 1.

The pressure transmitter

WHAT IT DOES	Measures the pressure in the pipe, through an isolating valve so that it can be removed without draining.
SPECIFICATION	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts
PLANNING ESTIMATE (EUR)	40 to 150. Not a quote.
USED IN	Kits H, F, W

On the pipe through an isolating valve, so that it can be removed without draining. A plumber does the wet connection. The pipe's pressure range decides the transmitter's range, and the top of that range, as the label says it, is what Greecon asks for. Wired to input 2.

The flow meter

WHAT IT DOES	Measures the flow in the pipe. Without it (or without the pump's running contact) the alert for a pump that runs with no water moving cannot be raised.
SPECIFICATION	4 to 20 mA output. Needed for "the pump is running but no water is flowing"
PLANNING ESTIMATE (EUR)	150 to 500. Not a quote.
USED IN	Optional in kits H, F, W

An electromagnetic or turbine meter in the pipe with a 4 to 20 mA output, installed with the straight run its manual asks for, and wired to input 3. Greecon asks for the top of its range, in litres a minute; where there is no flow meter the site ticks **Not fitted at this site**. Without it, or without the pump's running contact, "the pump is running but no water is flowing" cannot be raised, and the site should say so in its handover.

4.7 Farm sensors

A soil moisture probe and a temperature and humidity sensor that speak RS-485 Modbus RTU go on the same cable as everything else, each with its own unit number: soil probe 6, weather sensor 7 (set it before the probe is buried or the sensor mounted). They are not approved models: Greecon does not know their registers, and the box cannot recognise them on a network. So their address is entered by hand, and each reading is mapped by hand with **Choose the register**, from the maker's register table, which decides the register, the kind of number and the scale.

The soil moisture probe

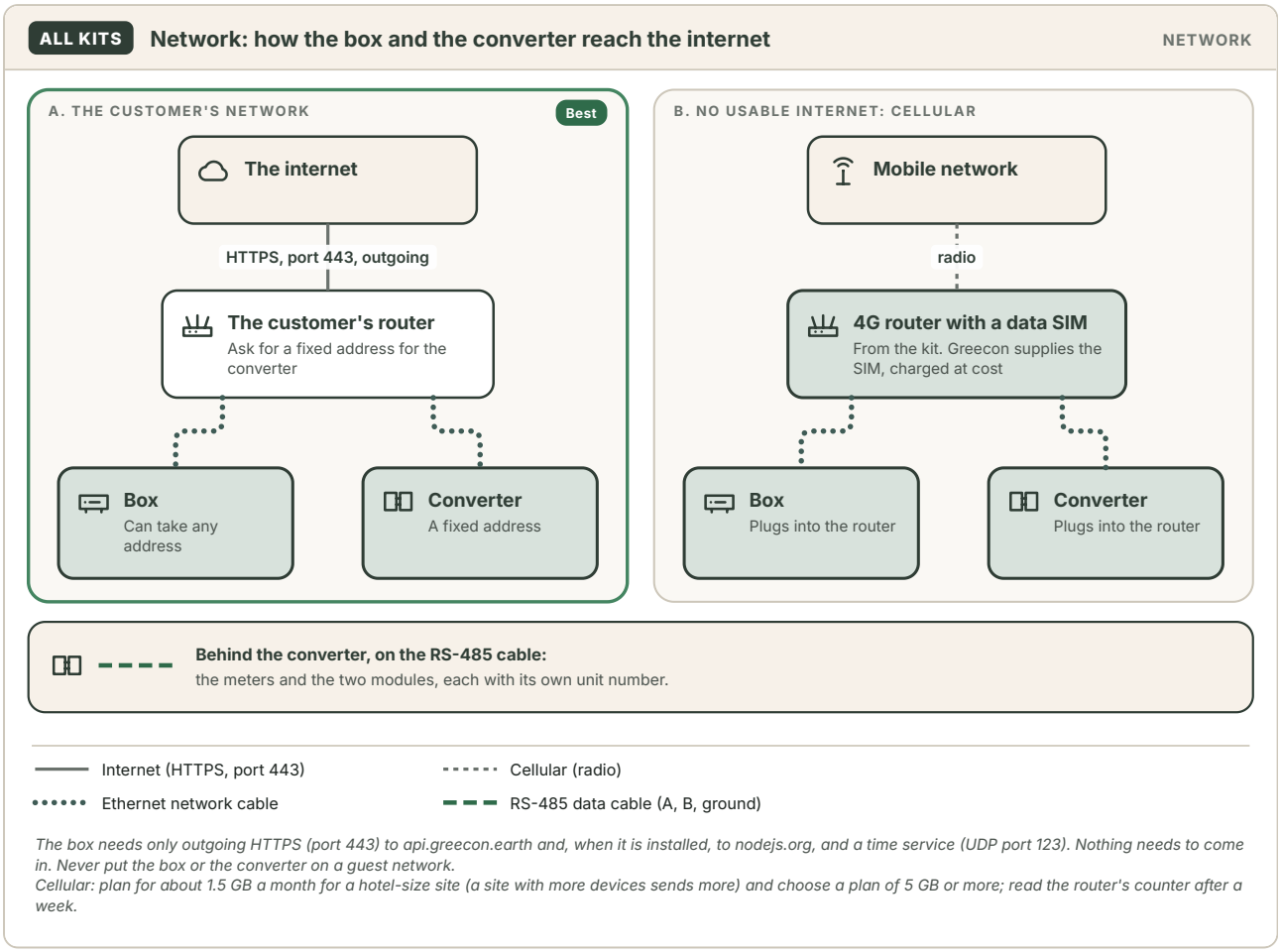
WHAT IT DOES	Measures soil moisture at the probe’s depth. Its depth and place are written down, because soil readings mean little without them. Unit 6, mapped by hand from the maker’s register table.
SPECIFICATION	RS-485 Modbus, with a published register table, one for each zone
PLANNING ESTIMATE (EUR)	40 to 120 each. Not a quote.
USED IN	Kit F

Write down the probe’s depth and where it is. Soil readings mean little without them, and they are only as good as the contact between the probe and the soil. Greecon gives no irrigation advice until the grower’s value for “start irrigating below” is entered: after the first week of readings, ask the grower what reading means “time to water”.

The temperature and humidity sensor

WHAT IT DOES	Measures air temperature and humidity at the field. Unit 7, mapped by hand from the maker’s register table.
SPECIFICATION	RS-485 Modbus, with a published register table
PLANNING ESTIMATE (EUR)	20 to 80. Not a quote.
USED IN	Kit F

4.8 The network



The box needs only outgoing HTTPS (port 443) to api.greecon.earth and, when it is installed, to nodejs.org, and a time service (UDP port 123). Nothing needs to come in. Never put the box or the converter on a guest network.
Cellular: plan for about 1.5 GB a month for a hotel-size site (a site with more devices sends more) and choose a plan of 5 GB or more; read the router's counter after a week.

Figure 4-5. The network: the customer's router (best), or a 4G router with a data SIM.

Situation	What to do
The customer's network	Box and converter on the router. Ask for a fixed address (a reservation) for the converter. The box needs only outgoing HTTPS (443) to `api.greecon.earth` (and to `nodejs.org` at install) and a time service (UDP 123). Nothing needs to come in. When someone presses **Find equipment**, the box looks at the site's own network for about a minute: it tries port 502 (or the port the person gives) on the private addresses of that network, such as 192.168.x.x and 10.x.x.x, only reads, and never writes to anything or looks outside the site's network. The install card has the words for the customer's network administrator.
Cellular	A 4G router with a data SIM; box and converter plug into its sockets. Plan for about 1.5 GB a month. That assumes a hotel-size site (four devices, seven readings) sending one request every ten seconds, about 5 KB each with its headers (about 1.3 GB), plus a heartbeat every 30 seconds (about 0.15 GB); a site with more devices sends more. It is an estimate: read the router's counter after a week and write the real figure down. Choose a plan of 5 GB or more.
No internet at all	Not sold. The box keeps about three days (for a site of up to about thirteen devices), which is for outages, not for a site that is never connected.

The cellular router

WHAT IT DOES	Gives the box and the converter an internet connection where the site has no usable one: a 4G router with a data SIM.
SPECIFICATION	4G, DIN rail, data SIM
PLANNING ESTIMATE (EUR)	150 to 200. Not a quote.
USED IN	Kit F; optional in kits H, S, W

Keep the box and the converter on separate addresses from the customer's own equipment where you can, and never on a guest network.

4.9 Power

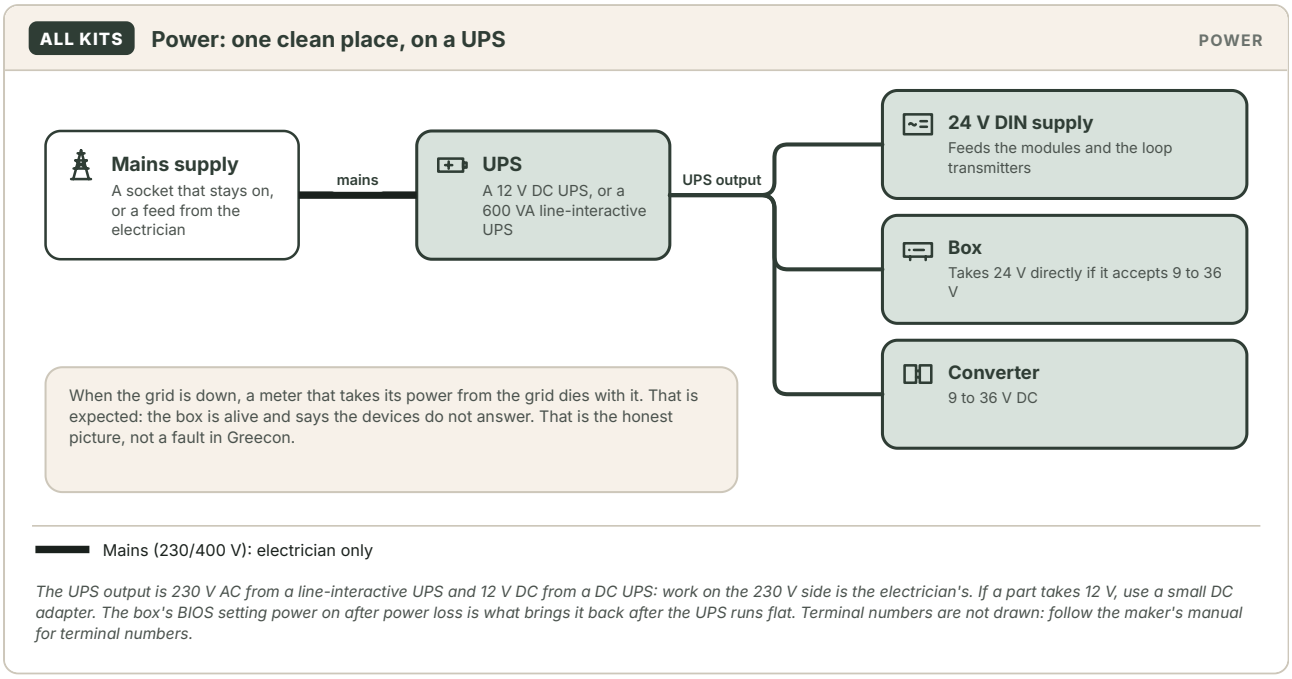


Figure 4-4. Power: everything from one clean place, on a UPS.

The 24 V power supply

WHAT IT DOES	Powers the two modules, the loop transmitters and the converter from one place.
SPECIFICATION	30 to 60 W
PLANNING ESTIMATE (EUR)	25 to 50. Not a quote.
USED IN	Kits H, S, F, W

The UPS

WHAT IT DOES	Keeps the box and the converter up through short power cuts and long enough to say that the grid is down.
SPECIFICATION	A 12 V DC UPS for the box and converter, or a 600 VA line-interactive UPS
PLANNING ESTIMATE (EUR)	60 to 150. Not a quote.
USED IN	Kits H, S, W

- One 24 V DIN supply feeds the two modules, the loop transmitters and, through a small DC adapter if the box takes 12 V, the converter. The box takes 24 V directly if it accepts 9 to 36 V.
- A UPS keeps the box and the converter up through short cuts and long enough to say that the grid is down. **When the grid is down, a meter powered from the grid dies with it.** That is expected: the box is alive and says the devices do not answer, which is the honest picture (“we cannot tell”), not a fault in Greecon.
- Auto power-on in the BIOS (section 4.3) is what brings the box back after the UPS runs flat.

4.10 Enclosure and labelling

The enclosure

WHAT IT DOES	Holds the converter, the modules, the supply, the fuses, the terminal blocks and, if there is room, the box.
SPECIFICATION	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands
PLANNING ESTIMATE (EUR)	80 to 150. Not a quote.
USED IN	Kits H, S, F, W

One IP54 box (IP65 if outside) holds the converter, the two modules (the 4 to 20 mA module and, where there is one, the pump contact module), the supply, the fuses, the terminal blocks and, if there is room, the box. Label everything: the unit number on each device (set at the office) and the settings on the converter, the wire numbers at both ends, and the site name and the date on a sticker inside the lid. **Never write a code or a password on the box.** Photograph the inside before closing.

4.11 The project manager's tools

The project manager brings:

- [ ] A laptop
- [ ] A USB-to-RS-485 adapter (to number a module or a probe, and to talk to one device alone)
- [ ] A loop calibrator (or a 4 to 20 mA source)
- [ ] A multimeter
- [ ] Screwdrivers (slotted and Phillips, small and medium)
- [ ] Wire strippers and a ferrule crimper
- [ ] A network cable tester
- [ ] Cable ties and labels
- [ ] A head torch
- [ ] Spare fuses and terminals
- [ ] A phone for photographs

You do not need a tool for work inside a live board: that is the electrician's.

4.12 Buying

1. Ask three suppliers for a quote. For the approved models (the two Eastron meters and the two Waveshare modules) the model is named, and a look-alike is not the same part; for everything else it is the specification, not the brand. Keep the quotes. (The Appendices of the Complete Deployment Guide (PDF 00) have a shopping list for each kit with room for three quotes.)

2. Buy one kit, bench-test it (the rehearsal), and fix the specification where it failed.
3. Then buy five. For each part keep: the make and model, the supplier, the date, the serial number or the batch, and the invoice. A spreadsheet is enough: the hardware register.
4. **Do not buy** a meter or a module that is not an approved model, a probe or a sensor whose register table you have not seen, a transmitter not rated for the fluid, or anything with a wireless link to the board that you cannot inspect.

Where the planning figures come from

Seen in October 2026, not quotes: Waveshare RS-485 to Ethernet converters, about EUR 25 to 60, and a look-alike at USD 45 (Electrokit, Botland); Eastron SDM630MCT V2 from about GBP 75 to 150 (Bimble Solar); a Waveshare 8-channel Modbus analog input module, about USD 25 to 36 (Waveshare); a Teltonika RUT241 4G router, about EUR 150 to 200 (Pricerunner); WIKA pressure transmitters, GBP 129 and up (RS); level transmitters from about USD 28 (unverified quality) to over GBP 900 (RS); fanless N100 mini PCs from about USD 112 in bulk to over USD 400 retail, with the real-time clock not stated by the listings: ask the vendor (Newegg). The Eastron register tables and defaults are in the maker's own manuals (the SDM630 Modbus protocol document and the SDM230 user manual V1.6): read the one that came with the meter before relying on this page, because versions differ. The planning figure for the pump contact module (30 to 60) is the earlier figure for a pump run input: it has not been checked against a shop price for the Waveshare Modbus RTU IO 8CH, and the registers of both Waveshare modules are from the maker's published tables and have not been tried on real hardware, so the first real kit is rehearsed on a desk and anything that differs is reported to Greecon.